SANATH S GOWDA

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SUMMARY

Software Developer and Computer Science graduate with hands-on experience in full-stack application development, scalable backend architectures, and high-performance distributed systems. Proficient in Python, Java, JavaScript, and SQL, with demonstrated experience engineering high-throughput REST and WebSocket APIs (FastAPI), interactive frontends (React Native), and cloud databases (Firestore, MySQL). State-level KSCST engineering grant recipient and 2x hackathon winner committed to developing clean, robust, and impactful production software.

EXPERIENCE

Android and Generative AI Intern | MindMatrix / CL Infotech Pvt. Ltd. **Feb 2026 – May 2026**

Mobile Architecture & Cloud Integration Bengaluru, India

- Engineered production mobile application modules using Kotlin and Jetpack Compose under MVVM architecture, ensuring smooth 60 FPS UI performance.
- Architected backend API integrations with Google Cloud and AI services, implementing asynchronous request pipelines and resilient error recovery.
- Designed scalable Cloud Firestore NoSQL schemas and Firebase Authentication workflows, authoring unit and integration test suites that reduced sprint bug rates by 35%.

PROJECTS

OmniCache & Guard — Distributed Semantic AI Gateway & WAF **2026 | Live App • GitHub**

Python, FastAPI, Hybrid Vector Embeddings, Cosine Similarity, Token-Bucket Rate Limiter, SQLite

- Engineered an OpenAI-compatible AI Gateway (/v1/chat/completions) featuring a Hybrid Semantic Vector Cache (word + subword n-grams) that cuts repeated inference latency from ~850 ms to ~2.4 ms (345x speedup) and reduces token costs by up to 65%.
- Built a Zero-Trust Security Firewall with 6 runtime-toggleable DLP PII scrubbers, Prompt-Injection/SQLi WAF rules (HTTP 403), O(1) Token-Bucket rate limiting, and a 3-state Circuit Breaker across multi-LLM fallback providers.

AutoRepo — Autonomous 4-Agent PR Reviewer & AST Security Linter **2026 | Live App • GitHub**

Python, Compiler AST (ast.NodeVisitor), FastAPI, Multi-Agent DAG, Pytest Synthesis, HMAC-SHA256

- Architected a 4-Agent Directed Acyclic Graph (SecOps SAST, Arch/AST Complexity, UnitTest Synthesizer, Lead Patch Critic) that parses Python Abstract Syntax Trees to compute McCabe Cyclomatic Complexity, loop Big-O depth, and N+1 queries.
- Automated OWASP Top-10 / CWE vulnerability detection (CWE-89 SQLi, CWE-798 Secrets, CWE-95 RCE), side-by-side security patch refactoring, and deterministic pytest unit test generation with signed GitHub Webhook verification.

FinSight — Multimodal Financial (SEC 10-K / Basel III) & Legal Hybrid RAG **2026 | Live App • GitHub**

Python, Sparse BM25Okapi, Dense Vector Search, Reciprocal Rank Fusion (RRF), Cross-Encoder, RAGAS

- Built an institutional Hybrid RAG engine combining Sparse Lexical BM25Okapi and Dense Vector Embeddings via Reciprocal Rank Fusion (RRF) and Cross-Encoder Reranking, achieving 97.8% Recall@3 on SEC 10-K and RBI regulatory filings.
- Implemented automated Markdown GAAP financial table extraction and a real-time RAGAS evaluation triad (Faithfulness, Context Precision, Answer Relevance) with verbatim [Doc-ID] citation grounding to eliminate hallucinations.

Upadha AI — Civil Services Mains OCR Evaluator & Interview Platform **2026 | Live App • GitHub**

Node.js, Gemini Vision OCR, Turso Cloud (libSQL), Capacitor Android SDK, SSE Streaming, PWA

- Architected a full-stack evaluation platform with real-time SSE streaming, multi-page handwriting OCR, and dual-mode Turso Cloud SQLite (libSQL) + local SQLite failover for 100% uptime; packaged for Android via Capacitor and deployed on Render.

TECHNICAL SKILLS

Languages:	Python, Java, JavaScript (ES6+), C, C++, SQL, HTML5/CSS3
AI & Backend Frameworks:	FastAPI, Node.js, Hybrid RAG (BM25 + RRF), Semantic Vector Caching, Multi-Agent DAGs, AST Static Analysis, React Native, Jetpack Compose
Databases & Cloud Tools:	SQLite, Turso Cloud (libSQL), PostgreSQL, MySQL, Cloud Firestore, Scikit-Learn, NumPy, Git, Linux, Render, Capacitor

ACHIEVEMENTS & CERTIFICATIONS

- **KSCST State Research Grant:** Awarded competitive research grant for Intelligent Fitness Assistant under 49th Series SPP.
- **Runner-up, Cyber Security Hackathon:** SIT (competed among 45+ collegiate engineering teams on secure system architecture).
- **Top Finalist, Ideathon:** BIT (recognized for assistive computer vision architecture and interactive product usability).
- **Software Testing (Elite Certification):** NPTEL — Verification, Validation, Unit Testing, TDD & Test Coverage.

EDUCATION

BGS College Of Engineering And Technology Bengaluru, India

B.E. in Computer Science & Engineering | **8.00 CGPA** **2022–2026**

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Software Engineering